KUM DONALSIEN AKWO

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PROFESSIONAL PROFILE

B.Tech Computer Engineering graduate specialising in Networks and Security at the College of Technology, University of Buea, with a strong passion for cybersecurity and cyberspace as a whole. Currently reading for a one year B.Eng in Computer Engineering at the Faculty of Engineering and Technology, University of Buea. Skilled in cybersecurity, network security engineering, AI/ML-based and hybrid intrusion detection systems and other network monitoring and auditing tools. Holds 2 years of hands-on technical experience in network installation, configuration, troubleshooting and security with Telnet Cameroon. Also designs, builds and deploys working software, with public open source projects in Python and JavaScript covering inactivity detection, scam detection, payment safeguards and language model evaluation. Writes tests as a matter of course and pays closest attention to how a system behaves when it fails. Proven leader as Program Delegate for the Computer Engineering Department, under the Computer Networks and Systems Maintenance option.

EDUCATION

FACULTY OF ENGINEERING AND TECHNOLOGY, UNIVERSITY OF BUEA Oct 2026 - Present

B.Eng Computer Engineering (One Year Programme) (In Progress)

COLLEGE OF TECHNOLOGY, UNIVERSITY OF BUEA Oct 2023 - Aug 2026

Bachelor of Technology (B.Tech) Computer Engineering (First Class Honours) (Completed)

3.73/4.0

GCE ADVANCED LEVEL June 2023 Session

General Certificate of Education (GCE) Board, Cameroon

- Biology, Chemistry, Pure Mathematics with Mechanics, Physics, Computer Science

SKILLS

- Cyber security (Kali Linux, Wireshark, Nmap, Snort, Suricata, IDS/IPS, Ethical Hacking, Penetration Testing, Fortigate, Data Security, Security Operations, Cyber Threat Intelligence, Social Engineering, Information Security)
- Networking (Cisco IOS, Cisco Switches & Routers, OSPF, BGP, IS-IS, RIP, Network Design, Wireless Networking, Network Architecture, IP Addressing)
- Cloud & Deployment (AWS EC2, S3, VPC, IAM, CloudFront, CloudWatch, Route 53, Lambda, EventBridge, SNS, Bedrock, VPC design, cloud networking, cloud security, Heroku, Render, Vercel, Cloudflare)
- Programming (Python, JavaScript, C, SQL, MySQL, PostgreSQL, SQLite)
- AI & Machine Learning (Scikit-learn, Keras/TensorFlow, Random Forest, XGBoost, CatBoost, Isolation Forest, Autoencoders, Deep Neural Networks, pandas, LLM Evaluation and Benchmarking, AI Agents)
- Frameworks & Tools (FastAPI, Flask, React, Tailwind CSS, Electron, Vite, Docker, Git, GitHub Actions, VirtualBox, MITRE ATT&CK)
- Testing & Reliability (pytest, building test suites from scratch, static checks in continuous integration, REST and webhook APIs, HMAC request signing, designing for failure modes)
- Operating Systems (Windows, Kali Linux)
- Productivity (Microsoft Office Suite)

EXPERIENCE

TELNET CM August 2024 - September 2024

Intern

- Configured enterprise networks with Cisco switches, routers and Fortigate firewalls, covering routing protocols (OSPF, BGP, IS-IS, RIP), wireless access points and secure remote access
- Deployed IP-based security surveillance systems and Starlink satellite internet infrastructure on client sites

TELNET CM October 2024 - Present

Support Staff

- Provides ongoing technical support and network engineering across client projects, including corporate network infrastructure deployments for enterprise and institutional clients
- Applies cybersecurity monitoring best practices and contributes to network security posture assessments

TELNET CM July 2025 - September 2025

Senior Intern & Intern Instructor

- Served as lead instructor and mentor for a new batch of junior interns, delivering training in network configuration, troubleshooting and cybersecurity best practices
- Supervised junior intern lab sessions covering Cisco IOS configuration, routing protocols and network security concepts

PROJECTS

Bastion IDS, A Hybrid Intrusion Detection System (2025-2026)

Four-layer hybrid intrusion detection system built as the final year capstone: a signature engine over 47,357 rules, a machine learning ensemble, a residual neural network and an anomaly detector, packaged as a desktop application for live network monitoring.

Technologies: *Python, FastAPI, React, Scikit-learn, XGBoost, CatBoost, Keras/TensorFlow, Wireshark, Snort*

Stillwatch, Inactivity Detection for People Living Alone (2026)

Tells somebody who left the house from somebody who stopped moving in it, by classifying each camera as a way out or an interior room and reading the sequence rather than a threshold. Learns each household's own rhythm hour by hour. A broken feed reports that it cannot tell rather than reporting calm, because a dead service and a quiet house look identical. Deployed and live under Apache 2.0, with 12 test suites and 5 static checks.

Technologies: *Python, Flask, PostgreSQL, Ring Partner API, AWS (Bedrock, SNS, CloudWatch, Lambda, EventBridge), Heroku, Cloudflare*

CheckAm, Multi-Agent Scam Check and Campaign Intelligence (2026)

Paste a suspicious message and find out whether it is a scam, built on the Stop Arnaque 237 dataset, with campaign-level intelligence for analysts who need to see past the single message. Shipped as a React progressive web app and a Telegram bot so it works on a cheap phone. Scored on a blind set rather than on the data it learned from and the blind figures are the ones published. 127 tests, Apache 2.0.

Technologies: *Python, React PWA, Multi-Agent Orchestration, NLP, Telegram Bot API, Docker*

Countersign, Payment Guardrails for AI Agents (2026)

Sits between an AI shopping assistant and a shop's PayPal account, so a stranger's email cannot talk the agent into moving money. Every refund or payout the agent proposes is checked against the shop's own rules before it leaves and anything outside them is refused with an explanation in plain words. Built for the PayPal AI Hackathon on the PayPal REST sandbox, under Apache 2.0.

Technologies: *Python, PayPal REST APIs, AI Agent Tooling, TypeScript, Render*

Ring Event Simulator, Testing Tool for the Ring API (2026)

A standalone generator of realistic multi-week household event streams in Ring's own shape, so inactivity detection can be built and tested without a house, a camera or API access. Zero dependencies, deterministic from a seed, with its own test suite, continuous integration and changelog. Published separately under MIT because the problem it solves belongs to anyone building on the Ring API without hardware.

Technologies: *Python, Synthetic Data Generation, GitHub Actions*

Stop Arnaque 237, Open Cameroon Scam Dataset (2026)

Created and released an open, labelled dataset of real scams in Cameroon, crowdsourced from the people who receive them and published with a Zenodo DOI under CC BY 4.0. Real scam messages sit beside the genuine mobile-money messages they imitate, in English, French and Pidgin, anonymized and labelled for scam-detection models.

Technologies: *Python, pandas, NLP, Data Anonymization, Hugging Face, Zenodo*

Does the Answer Survive Disagreement, LLM Sycophancy Benchmark (2026)

Designed and published an open benchmark measuring whether a model keeps a correct answer when a user asserts a wrong one, with a control arm that separates caving from sensibly updating on evidence. Ran it across 12 models from six labs on Kaggle Benchmarks and published a technical write-up on DEV.

Technologies: *Python, pandas, Data Visualization, Kaggle Benchmarks*

FairGlass, Proof of Fair AI Hiring (2026)

Built on the Midnight blockchain during an MLH hackathon. A Compact smart contract proves that an AI hiring decision followed an agreed fairness policy, issuing a verifiable receipt when the rules hold and refusing one when a forbidden attribute changes the outcome, all without exposing candidate data. Contributed the backend, scoring logic and the contract bridge.

Technologies: *Python, Flask, Midnight (Compact smart contracts), Cryptographic Commitments*

Evaluation and Recommendation of Network Monitoring Tools (Aug - Sep 2024)

Evaluated industry-standard IDS/IPS and network monitoring tools to identify and prevent cyber threats. Produced formal analysis and recommendations report to enhance organisational security posture.

Technologies: *Wireshark, Snort, Suricata, Kali Linux, Nmap, Security Onion, Kitsune*

PUBLICATIONS

Tatsachou Ngningaye D., **Kum Donalsien A.**, Gamom Ngounou Ewo R. C., and Boum Alexandre T. "Bastion IDS: Layer Contribution, Threshold Sensitivity, and Consensus Rule Design in a Deployed Hybrid Intrusion Detection System." Journal of Cyber Security Technology (Taylor & Francis Group), 2026. Submitted.

Kum, D. A. (2026). Stop Arnaque 237: Cameroon Scam Reports Dataset (Version 1.0) [Data set]. Zenodo. https://doi.org/10.5281/zenodo.23287618

LANGUAGE

- English (Fluent), French (Basic)

ACTIVITIES

- 237HackFest Capture the Flag (CTF) Competition, organised by Hack The Box Cameroon, gained hands-on experience solving real-world cybersecurity challenges
- Major League Hacking in partnership with the Midnight network, 48-hour hackathon building data protective, privacy-first applications. Developing dApps and tools with ZK technology that put users back in control of their personal data.
- Program Delegate, Computer Engineering Department, Option: Computer Networks and Systems Maintenance, University of Buea, key liaison between students and university administration
- PayPal AI Hackathon (November 2026), built and submitted Countersign, payment guardrails that stop a prompt-injected AI agent from moving a shop's money
- Amazon Build Ship Shape Hackathon (October 2026), built and submitted Stillwatch, a deployed inactivity detection service for people living alone, on the Ring Partner API
- Open Agent Hackathon, GenAI Works (October 2026), built and submitted CheckAm, a multi-agent scam checker for Cameroonian users, with a separate entry for the Zetaris demo challenge
- Kaggle Benchmarks (October 2026), published a 112 question benchmark run across 12 language models from six labs, with the method and the data open so the numbers can be checked
- Independent security research on Intigriti, working private bug bounty programmes since October 2026
- Major League Hacking Global Hack Week: Data (September 2026)

SOFT SKILLS

- Leadership
- Teamwork
- Critical Thinking
- Problem Solving
- Effective Communication
- Adaptability
- Time Management
- Attention to Detail
- Resilience

CERTIFICATIONS

- Certified in Cybersecurity (CC), ISC2 (September 2026, valid to September 2029)
- Fortinet Certified Fundamentals (FCF) in Cybersecurity, Fortinet (July 2026, valid to July 2028)
- Cisco Introduction to Cybersecurity, Cisco Networking Academy (September 2025)
- Cisco Networking Academy Capture the Flag 2026, Cisco (May 2026)
- Cyber Crime, Humanitarian Leadership Academy (July 2024)
- Cyber Security, Humanitarian Leadership Academy (July 2024)

ACHIEVEMENTS & AWARDS

- Academic Excellence Recognition, consistently maintaining an excellent CGPA throughout the degree programme
- Leadership skills award from the College of Technology for demonstrating outstanding leadership capabilities in school
- Best Intern Award, Recognised by TELNET CAMEROON management as best intern for the August-September 2024 batch